Euiseok Han

hanes1207@kaist.ac.kr hanes1207.github.io linkedin.com/in/hanes1207

Research Interests

- Uncertainty-Aware Decision Making for Autonomous Systems: Multimodal Perception, State Estimation, Robot Policies, 3D Scene Understanding

Education

- **Korea Advanced Institute of Science and Technology, Daejeon** *Mar. 2020 – Aug. 2027 (Expected)*
  B.S. in Electrical Engineering and Computer Science (Double Major) GPA: 4.02/4.3
  – *Coursework*: Machine Learning for 3D Data, Kalman Filtering (graduate), Mathematical Physics

- **University of Texas at Austin** *Aug. 2025 – Dec. 2025*
  Exchange Program | University Honors (Fall 2025) GPA: 4.0/4.0
  – *Coursework*: Digital Signal Processing, Classical Electrodynamics, Quantum Physics

Publications

[1] **SenseFuse: Label-Free Fusion of Image and Shape Encoders for Open-Vocabulary 3D Instance Segmentation** *Under review*
Euiseok Han, Tri Ton, Hwanhee Kim, Seungyeon Ryu, Chang D. Yoo
IEEE Int. Conf. Robot. Autom. (ICRA) 2027 Project page arXiv Code Real-robot demo

[2] **Reference-Filter-Driven Transition Probabilities for IMM-Based Satellite Maneuver Detection** *Preprint*
Euiseok Han, Sangheon Choi, Seung-Hyun Kong
arXiv preprint Project page arXiv Interactive demo

Research Experience

- **Artificial Intelligence and Machine Learning Lab, KAIST** *Jan. 2025 – Feb. 2025, Jan. 2026 – Present*
  Undergraduate Researcher | Advisor: Prof. Chang D. Yoo Daejeon, KR
  – URP: *From Point Cloud to Actions: 3D Structural Keypoint-Conditioned Hierarchical Robot Policy Learning*
  – Revealed that frozen open-vocabulary 2D image and 3D shape encoders fail on largely disjoint instances and are weakly correlated (Spearman ρ 0.08–0.23, 5.9% error overlap), versus 0.68 and 41.4% for a 2D–2D pair [1]
  – Devised a 2D–3D fusion framework with a closed-form, label-free scene-level weight via sensitivity maximization (3.5 ms), achieving a median gain of +7.1 points and 93% of the oracle-weight gain
  – Implemented Gymnasium and joystick teleoperation interfaces for a UR5 robot; trained a diffusion policy on the collected demonstrations

- **Cho Chun Shik Graduate School of Mobility, KAIST** *Mar. 2026 – Present*
  Undergraduate Researcher | Advisor: Prof. Seung-Hyun Kong Daejeon, KR
  – Originated from a graduate Kalman filtering course project on a J_2-Hamiltonian-driven IMM
  – Identified a feedback loop in adaptive-TPM IMMs: statistics computed inside the IMM depend on the TPM through mixing, which sustains false alarms or suppresses detections
  – Proposed RD-IMM, driving the transitions from the NIS of a reference coast filter outside the IMM [2]
  – Detected 98% of 1 m/s north–south (out-of-plane) burns missed by a fixed-TPM IMM, with few false alarms

- **NC Lab, KAIST** *Mar. 2025 – Aug. 2025*
  Undergraduate Researcher | Advisor: Prof. Junehwa Song Daejeon, KR
  – *Socially-appropriate Use of Noise-making Everyday Appliances in High-density Urban Housing*
  – Participated in experiment design and conducted a preliminary experiment using Zigbee modules

•Control and Intelligent Systems Lab, KAIST *Jul. 2024 – Aug. 2024*

Undergraduate Researcher | Advisor: Prof. Heejin Ahn Daejeon, KR

– Reproduced Constrained Policy Optimization (ICML 2017) baselines in MuJoCo (safe RL)

Selected Projects

•LLM-Guided Autonomous Robots *Mar. 2026 – Jun. 2026*

EE40078 Introduction to Multi-disciplinary Robotics, School of Electrical Engineering

– LLM-guided autonomous mobile delivery robot; RTAB-Map localization, path planning, and object pick-up
– Built the RGB-D grasping pipeline with a Kalman-filtered target estimate and the ROS mission orchestrator

•PintOS-KAIST *Sep. 2024 – Dec. 2024*

CS330 Operating Systems, School of Computing

– Implemented kernel modules for process scheduling, syscall, memory management of an educational x86-64 OS

Honors and Awards

•Korea-U.S. Advanced Industries Youth Exchange Scholarship *Aug. 2025*

$9,000 for advanced-industry exchange study in the U.S.

Other Experience

•Republic of Korea Air Force (mandatory military service) *Apr. 2022 – Jan. 2024*

Information System Operator, Staff Sergeant, KAMD Operations Center Osan Air Base, KR

– Operated 24/7 real-time data processing servers for the Korea Air and Missile Defense center

Technical Skills and Interests

Programming Languages: C, C++, Python
Libraries: ROS, NumPy, PyTorch, Gymnasium, MuJoCo
Miscellaneous tools: MATLAB, LaTeX, Docker
Languages: Korean (native), English (TOEFL iBT 5.5/6)
Extracurricular: STROKE (Tennis Club)